The Mathematical Association of Victoria

Trial Examination 2025

SPECIALIST MATHEMATICS

Written Examination 2 SOLUTIONS

SECTION A- Multiple-choice questions

1	B	11	D
2	D	12	C
3	D	13	B
4	A	14	D
5	C	15	C
6	D	16	A
7	A	17	B
8	B	18	A
9	C	19	C
10	A	20	A

Question 1 Answer B

For statement, if P then Q

Contrapositive statement, if Q' then P'

Here, $P = n^2 - 2n + 7$ is even $\quad P' = n^2 - 2n + 7$ is odd

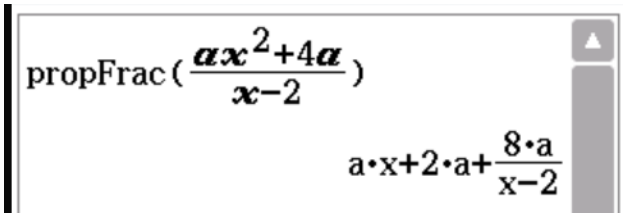
$Q = n$ is even $\quad Q' = n$ is odd

So, the contrapositive statement is 'For all integers, if n is odd, then $n^2 - 2n + 7$ is odd.'

Question 2 Answer D

$f(x) = \dfrac{ax^2 + 4a}{x - 2}$, asymptotes at $x = 2$ and $y = ax + 2a$

$f(x) = \dfrac{a\left(x^2 + 4\right)}{x - 2}$ gives vertical asymptote at $x = 2$



gives oblique asymptote at $y = ax + 2a$

Question 3 Answer D

Consider $y = -\sec(n\pi x + \pi) = -\sec(A)$ where $A = n\pi x + \pi$

First positive local maximum point in $y = -\sec(A)$ is $A = 2\pi$

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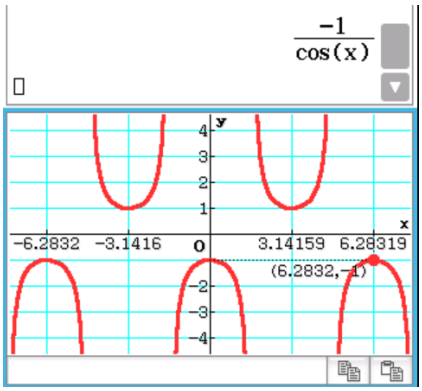


Let $A = n\pi x + \pi = 2\pi$

Gives $x = \dfrac{1}{n}$

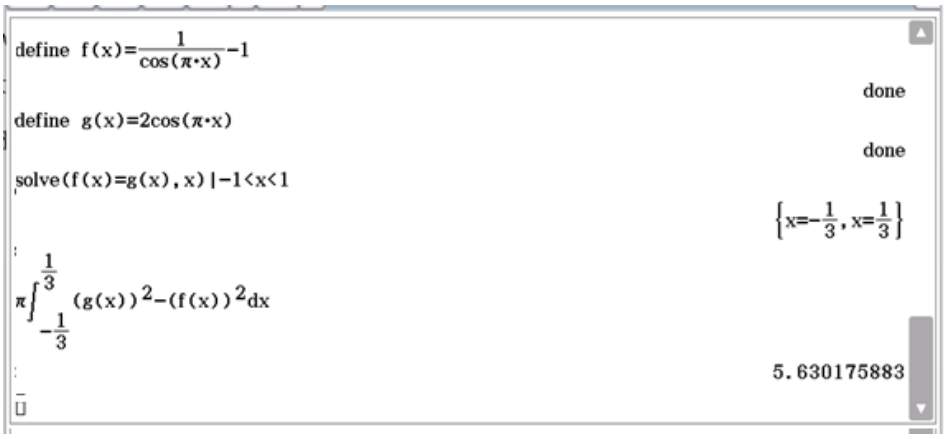
Question 4 Answer A

$f(x) = \sec(\pi x) - 1$ and $g(x) = 2\cos(\pi x)$.

Solve $\sec(\pi x) - 1 = 2\cos(\pi x)$ within domain $x \in (-1,1)$.

Points of intersection $x = \pm\dfrac{1}{3}$

Volume of the solid of revolution $= \pi \displaystyle\int_{-\frac{1}{3}}^{\frac{1}{3}} \left((g(x))^2 - (f(x))^2 \right) = 5.63017...$



Question 5 Answer C

$x = \sin(t) + 1$ and $y = 2t + \cos(t)$, where $t \in [0, \pi]$.

Length $= \displaystyle\int_0^{\pi} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$ where $\dfrac{dx}{dt} = \cos(t)$ and $\dfrac{dy}{dt} = 2 - \sin(t)$

$$l = \int_0^{\pi} \sqrt{(\cos(t))^2 + (2 - \sin(t))^2}\, dt$$

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$$=\int_0^{\pi}\sqrt{\cos^2(t)+4-4\sin(t)+\sin^2(t)}dt$$

But $\cos^2(t)+\sin^2(t)=1$

$$=\int_0^{\pi}\sqrt{5-4\sin(t)}dt$$

Since $0\le\sin(t)\le 1$ for $t\in[0,\pi]$ so that $5-4\sin(t)>0$

Question 6 Answer D

$|z-2+2i|\le 4$

$\Rightarrow |x+yi-2+2i|\le 4$

consider $|(x-2)+(y+2)i|=4$

$\sqrt{(x-2)^2+(y+2)^2}=4$

Gives $(x-2)^2+(y+2)^2=16$

Shading inside and including circle becomes $(x-2)^2+(y+2)^2\le 16$

Question 7 Answer A

Given $x(t)=\cos^2(t)+1$ and $y(t)=\tan^2(t)$ where $t\ge 0$.

Gradient at $t=\frac{\pi}{4}$.

<u>Method 1 using chain rule</u>

$x(t)=\cos^2(t)+1$

$\therefore \frac{dx}{dt}=-2\cos(t)\sin(t)=-\sin(2t)$

$y(t)=\tan^2(t)$

$\therefore \frac{dy}{dt}=2\tan(t)\sec^2(t)$

Chain Rule $\frac{dy}{dx}=\frac{dy}{dt}\times\frac{dt}{dx}$

$\frac{dy}{dx}=\frac{2\tan(t)\sec^2(t)}{-\sin(2t)}$

at $t=\frac{\pi}{4}$

$\frac{dy}{dx}=\frac{2\times 1\times 2}{-1}=-4$

<u>Method 2 using Cartesian equation</u>

$x-1=\cos^2(t)\therefore \sec^2(t)=\frac{1}{x-1}$

Use $\sec^2(t)=1+\tan^2(t)$

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$$\frac{1}{x-1}=1+y$$

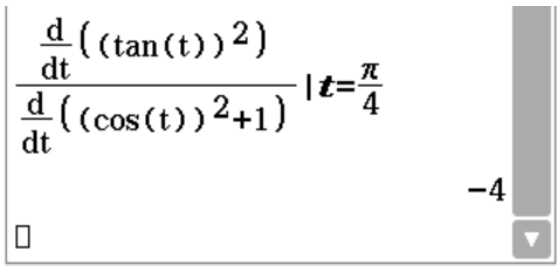
$$\therefore y=\frac{1}{x-1}-1$$

$$\frac{dy}{dx}=\frac{-1}{(x-1)^2}$$

At $t=\frac{\pi}{4}, x(t)=\cos^2(t)+1 \therefore x=\cos^2\left(\frac{\pi}{4}\right)+1 \Rightarrow x=\frac{3}{2}$

So $\frac{dy}{dx}=\frac{-1}{(x-1)^2}=\frac{-1}{\left(\frac{3}{2}-1\right)^2}=-4$

<u>Method 3 using CAS</u>



Question 8 Answer B

$$\int_0^2\left((2x+1)^2\sqrt{1+4x}\right)dx$$

Let $u=1+4x \therefore \frac{du}{dx}=4$

Also $x=\frac{u-1}{4}$

When $x=0, u=1$ and $x=2, u=9$

$$\frac{1}{4}\int_1^9\left(\left(2\left(\frac{u-1}{4}\right)+1\right)^2\sqrt{u}\right)du$$

$$\frac{1}{4}\int_1^9\left(\left(\left(\frac{u}{2}-\frac{1}{2}\right)+1\right)^2\sqrt{u}\right)du$$

$$=\frac{1}{4}\int_1^9\left(\left(\frac{u}{2}+\frac{1}{2}\right)^2\sqrt{u}\right)du$$

$$=\frac{1}{16}\int_1^9\left(\left(u^2+2u+1\right)u^{\frac{1}{2}}\right)du$$

$$=\frac{1}{16}\int_1^9\left(u^{\frac{5}{2}}+2u^{\frac{3}{2}}+u^{\frac{1}{2}}\right)du$$

Question 9 Answer C

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$a = x + x^2$

$\therefore \dfrac{d\left(\frac{1}{2}v^2\right)}{dx} = x + x^2$

gives $\frac{1}{2}v^2 = \int\left(x + x^2\right)dx$

$\frac{1}{2}v^2 = \frac{x^2}{2} + \frac{x^3}{3} + c$

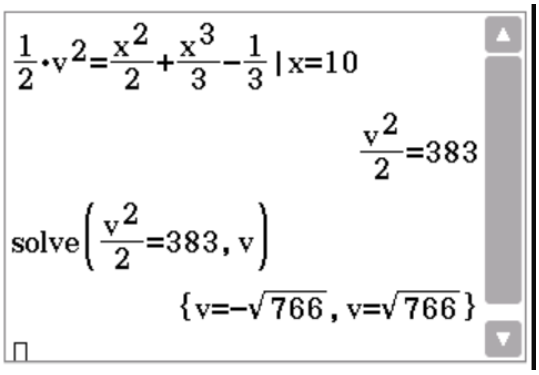
$v = 1, x = 1 \therefore c = -\frac{1}{3}$

$\frac{1}{2}v^2 = \frac{x^2}{2} + \frac{x^3}{3} - \frac{1}{3}$

$\therefore v^2 = x^2 + \frac{2x^3}{3} - \frac{1}{3}$

when $x = 10, v = \pm\sqrt{766}$

reject negative



Question 10 Answer A

Slope field defined only in the first and third quadrants?

$\frac{dy}{dx} = \log_e\left(\frac{x}{10y}\right)$

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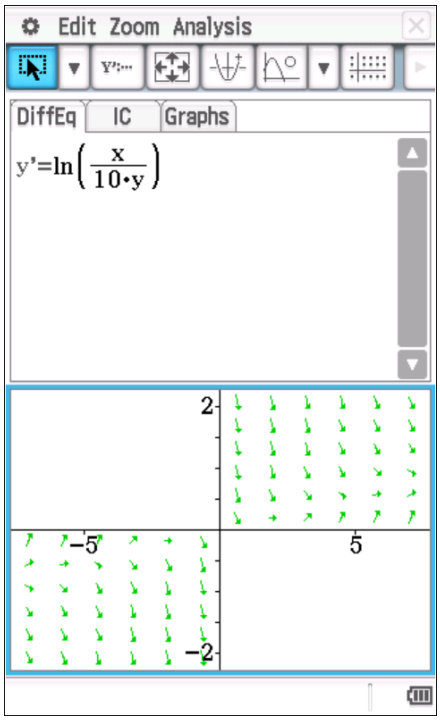

Question 11 Answer D

$$\frac{dy}{dx} = \frac{y-1}{x^2+1}$$

$$\Rightarrow g(x,y) = \frac{y-1}{x^2+1}$$

$$y_1 = y_0 + h \times g(x_0, y_0) = 2 + h \times \frac{2-1}{0+1} = 2 + h \qquad x_1 = x_0 + h = h$$

$$y_2 = y_1 + h \times g(x_1, y_1) = 2 + h + h \times \frac{2+h-1}{h^2+1} = 2 + h + h \times \frac{h+1}{h^2+1} \qquad x_2 = x_2 + h = 2h$$

Question 12 Answer C

$\underset{\sim}{a} = 3\underset{\sim}{i} + 4n\underset{\sim}{j} + \underset{\sim}{k}$, $\underset{\sim}{b} = -\underset{\sim}{i} - 2\underset{\sim}{j} + 2n\underset{\sim}{k}$ and $\underset{\sim}{c} = 2\underset{\sim}{i} + \underset{\sim}{k}$

Assume **linear dependence**

$\underset{\sim}{a} = p\underset{\sim}{b} + q\underset{\sim}{c}$

$\Rightarrow 3\underset{\sim}{i} + 4n\underset{\sim}{j} + \underset{\sim}{k} = p\left(-\underset{\sim}{i} - 2\underset{\sim}{j} + 2n\underset{\sim}{k}\right) + q\left(2\underset{\sim}{i} + \underset{\sim}{k}\right)$

$\underset{\sim}{i}$	$\underset{\sim}{j}$	$\underset{\sim}{k}$
$\Rightarrow 3 = -p + 2q$	$\Rightarrow 4n = -2p$	$\Rightarrow 1 = 2np + q$

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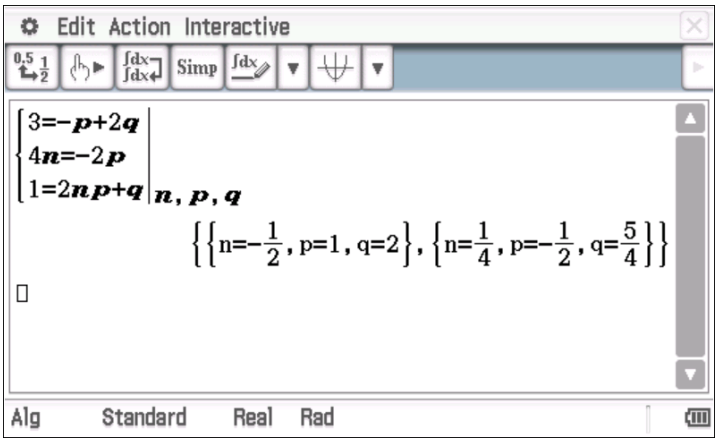


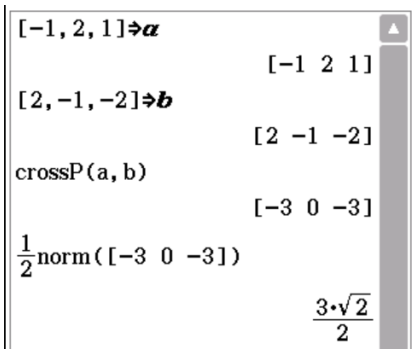
For **linearly independent**

$$n \in R \setminus \left\{-\frac{1}{2}, \frac{1}{4}\right\}, p \in R \setminus \left\{1, -\frac{1}{2}\right\}, q \in R \setminus \left\{2, \frac{5}{4}\right\}$$

Question 13 Answer B

A parallelogram OABC with $\vec{OA} = -\underset{\sim}{i} + 2\underset{\sim}{j} + \underset{\sim}{k}$ and $\vec{OC} = 2\underset{\sim}{i} - \underset{\sim}{j} - 2\underset{\sim}{k}$.

Area of triangle ABC is $\frac{1}{2}\left|\vec{OA} \times \vec{OC}\right| = \frac{3\sqrt{2}}{2}$



Question 14 Answer D

$\underset{\sim}{a} = m\underset{\sim}{i} - 2\underset{\sim}{j}$ and $\underset{\sim}{b} = 2\underset{\sim}{i} - m\underset{\sim}{j}$, , The obtuse angle between $\underset{\sim}{a}$ and $\underset{\sim}{b}$ is 150^0

$$\underset{\sim}{a} \cdot \underset{\sim}{b} = |\underset{\sim}{a}||\underset{\sim}{b}|\cos(\theta)$$

$$\cos(\theta) = \frac{\underset{\sim}{a} \cdot \underset{\sim}{b}}{|\underset{\sim}{a}||\underset{\sim}{b}|} = \frac{4m}{m^2 + 4}$$

As $m \in R^+, \frac{4m}{m^2 + 4} > 0$

So, $\frac{4m}{m^2 + 4} = \cos\left(150^o\right)$

$$\Rightarrow m = -2\sqrt{3}, -\frac{2\sqrt{3}}{3}$$

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```
[m -2]⇒a
                                  [m -2]
[2 -m]⇒b
                                  [2 -m]
dotP(a, b)
norm(a)norm(b)
                                  4·m
                                  m^2+4
solve(4·m/(m^2+4)=cos(150), m)
                  {m=-2·√3, m=(-2·√3)/3}
```

Question 15 Answer C

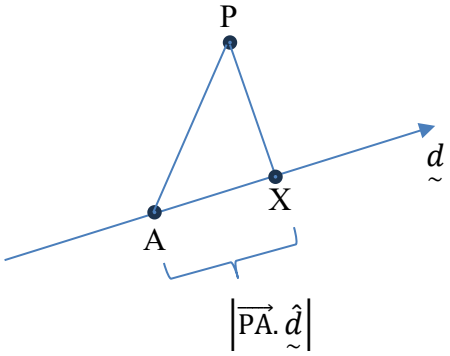
$\frac{x-1}{2}=y+1=\frac{-z+2}{2}$. Shortest distance from the point $(3,-2,0)$ to the line is

$\frac{x-1}{2}=y+1=\frac{-z+2}{2}=t$

$\Rightarrow x=1+2t$

$\Rightarrow y=-1+t$

$\Rightarrow z=2-2t$



Direction vector of line, $\underset{\sim}{d}=2\underset{\sim}{i}+\underset{\sim}{j}-2\underset{\sim}{k}$

Point on line, $A(1,-1,2)$

Distance from point, $P(3,-2,0)$

$\overrightarrow{AP}=2\underset{\sim}{i}-\underset{\sim}{j}-2\underset{\sim}{k}$

$\left|\overrightarrow{AP}\right|=\sqrt{2^2+1^2+2^2}=3$

Magnitude of scalar projection of $\overrightarrow{AP}$ towards $\underset{\sim}{d}$, $\left|\overrightarrow{AX}\right|=\left|\overrightarrow{AP}.\hat{d}\right|=\frac{7}{3}$

$\left|\overrightarrow{PX}\right|=\sqrt{\left|\overrightarrow{AP}\right|^2-\left|\overrightarrow{AX}\right|^2}=\sqrt{3^2-\left(\frac{7}{3}\right)^2}=\frac{4\sqrt{2}}{3}$

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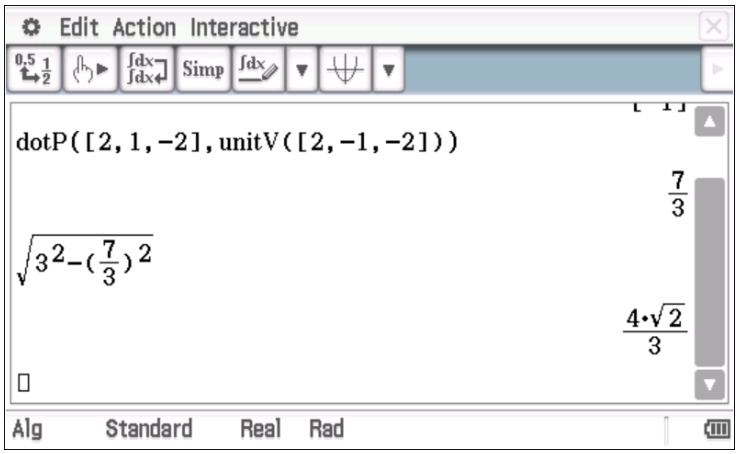


Question 16 Answer A

$\underset{\sim}{a} = m\underset{\sim}{i} + \underset{\sim}{j} + 2\underset{\sim}{k}$, $\underset{\sim}{b} = 2\underset{\sim}{i} + \underset{\sim}{j} - m\underset{\sim}{k}$ and $\underset{\sim}{c} = -3\underset{\sim}{i} - \underset{\sim}{j} + \underset{\sim}{k}$. Vectors $\underset{\sim}{a}$, $\underset{\sim}{b}$ and $\underset{\sim}{c}$ lie on the same plane.

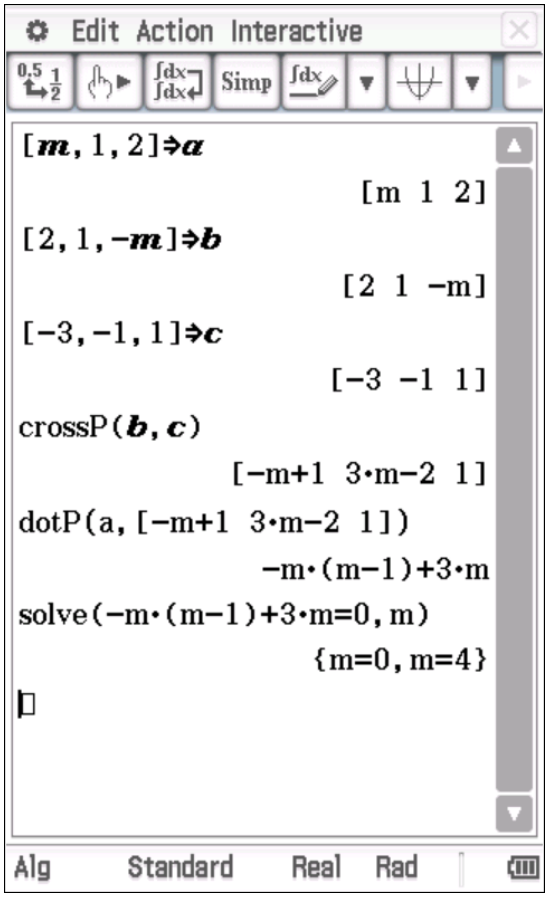
Normal vector to the plane, $\underset{\sim}{n} = \underset{\sim}{b} \times \underset{\sim}{c} = (-m+1)\underset{\sim}{i} + (3m-2)\underset{\sim}{j} + \underset{\sim}{k}$

Vector $\underset{\sim}{a}$ and $\underset{\sim}{n}$ are perpendicular. So,

$\underset{\sim}{n}.\underset{\sim}{a} = 0$

$\Rightarrow m(-m+1) + (3m-2) + 2 = 0$

$\Rightarrow m = 0, m = 4$



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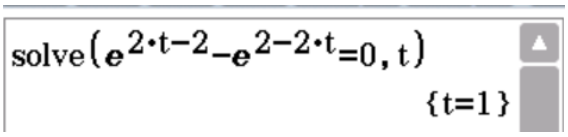
Question 17 Answer B

Velocity $\dot{\underset{\sim}{r}}(t) = e^{t-1}\underset{\sim}{i} + e^{1-t}\underset{\sim}{j},\ t \geq 0.$

Acceleration $\ddot{\underset{\sim}{r}}(t) = e^{t-1}\underset{\sim}{i} - e^{1-t}\underset{\sim}{j}$

For velocity and acceleration to be perpendicular,
$\dot{\underset{\sim}{r}} \cdot \ddot{\underset{\sim}{r}}(t) = 0 \Rightarrow e^{2t-2} - e^{2-2t} = 0 \Rightarrow t = 1$



Question 18 Answer A

Vector projection of vector $\underset{\sim}{a}$ in the direction of vector $\underset{\sim}{b}$,

$$= \left(\underset{\sim}{a}.\hat{\underset{\sim}{b}}\right)\hat{\underset{\sim}{b}} = \left(\underset{\sim}{a}.\frac{\underset{\sim}{b}}{|\underset{\sim}{b}|}\right)\frac{\underset{\sim}{b}}{|\underset{\sim}{b}|} = \frac{(\underset{\sim}{a}.\underset{\sim}{b})}{|\underset{\sim}{b}|^2}\underset{\sim}{b} = \frac{(a_1 \times b_1 + a_2 \times b_2 + a_3 \times b_3)}{(b_1)^2 + (b_2)^2 + (b_3)^2}\left(b_1\underset{\sim}{i} + b_2\underset{\sim}{j} + b_3\underset{\sim}{k}\right)$$

$$x = (a_1 \times b_1 + a_2 \times b_2 + a_3 \times b_3) \times \frac{b_1}{(b_1)^2 + (b_2)^2 + (b_3)^2}$$

$$y = (a_1 \times b_1 + a_2 \times b_2 + a_3 \times b_3) \times \frac{b_2}{(b_1)^2 + (b_2)^2 + (b_3)^2}$$

$$z = (a_1 \times b_1 + a_2 \times b_2 + a_3 \times b_3) \times \frac{b_3}{(b_1)^2 + (b_2)^2 + (b_3)^2}$$

So, $m = (a_1 \times b_1 + a_2 \times b_2 + a_3 \times b_3)$

Question 19 Answer C

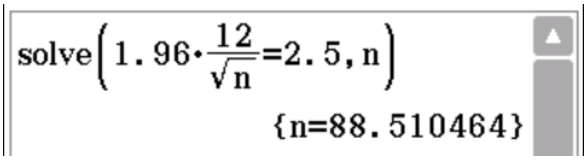
n persons age at which they got married. 95% confidence interval is $(28.5, 33.5)$ years.
Sstandard deviation is 12 years.

Error margin, $M = \dfrac{33.5 - 28.5}{2} = 2.5$

So, $M = 1.96 \times \dfrac{\sigma}{\sqrt{n}} = 2.5$

$\Rightarrow 1.96 \times \dfrac{12}{\sqrt{n}} = 2.5$

$\Rightarrow n = 88.51 \approx 89$



Question 20 Answer A

Type II error: Null hypotheis is false but hypothesis testing fails to reject null hypothesis.

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In this example the null hypothesis being false means “the company actually pays more salary”, and failing to reject null hypothesis means “test concludes that company pay is unchanged, i.e. does not pay more salary”.

The company pays more salary, but the test concludes that the company does not.

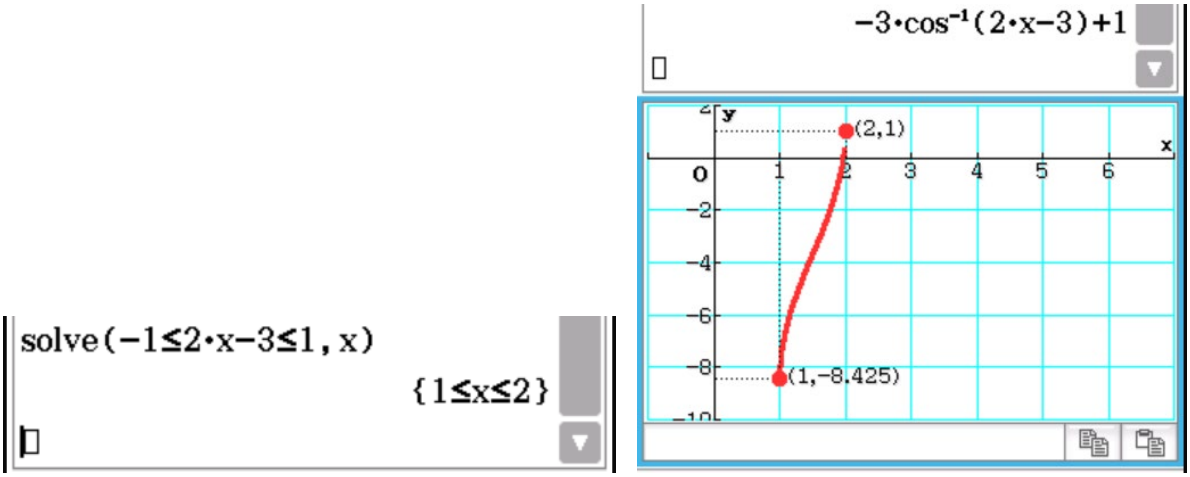
SECTION B

Question 1 (10 marks)

$f(x) = -3\arccos(2x-3)+1$

a. $-1 \le 2x-3 \le 1$ gives domain: $[1,2]$

range of the graph of f : $[-3\pi+1,1]$



Graph of $y = f(x)$. **3A**

Endpoints $(1,-3\pi+1)$ and $(2,1)$

Point of inflection: by inspection from graph $\left(\frac{3}{2},\frac{-3\pi+2}{2}\right)$

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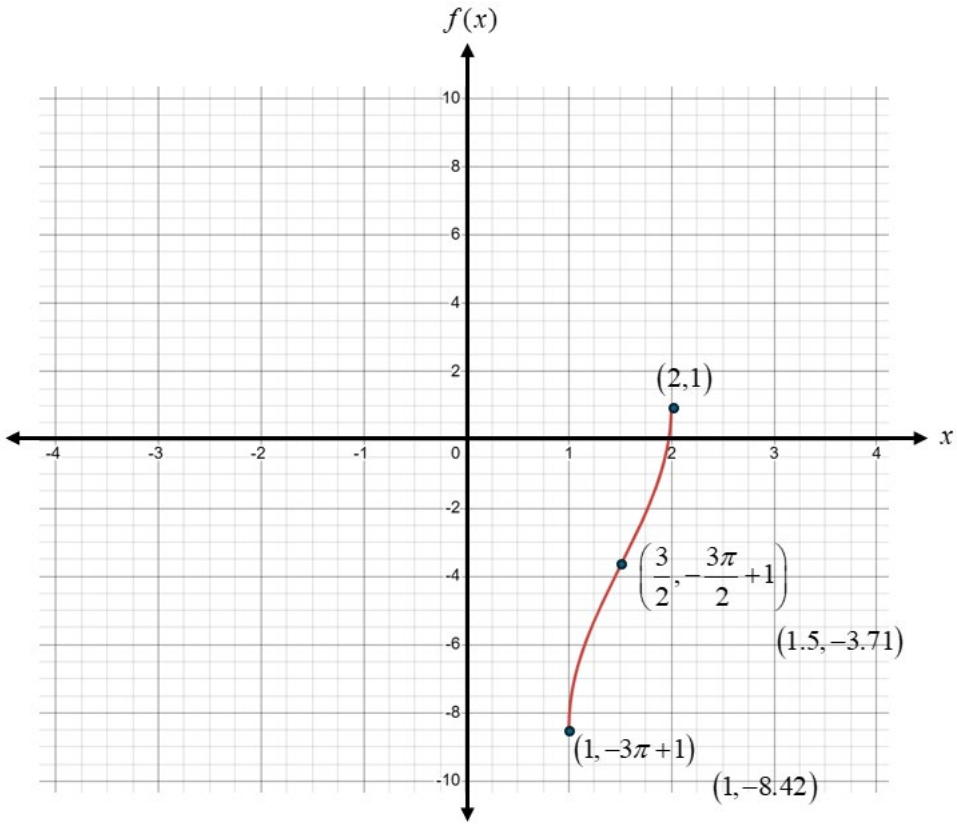
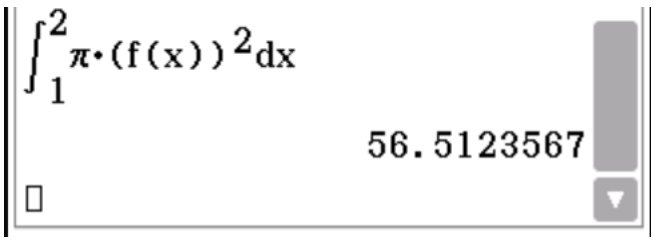


b. f rotated around the $x-$axis.

i. Volume $=\pi\int_1^2 (f(x))^2\,dx$. **1A**

ii. $\pi\int_1^2 (f(x))^2\,dx = 56.51$ **1A**



$g(x) = a\arccos(bx+c)+d$

c. **i.** point of inflection on the graph of g where $a>0, b\neq 0$

$-1\le bx+c\le 1$ gives domain: $\left[\frac{-1-c}{b},\frac{1-c}{b}\right]$ **1M**

range of the graph of f : $[d, a\pi+d]$

Point of inflection: $\left(\frac{-c}{b},\frac{a\pi+2d}{2}\right)$, where $a>0, b\neq 0$ **1A**

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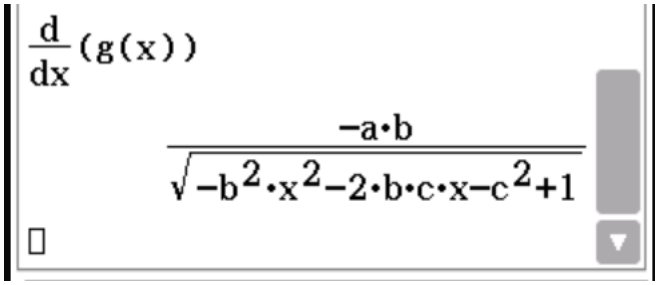
ii. $g(x) = a\arccos(bx + c) + d$

$$\therefore g'(x) = \frac{-ab}{\sqrt{-b^2x^2 - 2bcx - c^2 + 1}}$$

No solution for x for $\dfrac{-ab}{\sqrt{-b^2x^2 - 2bcx - c^2 + 1}} = 0$ **1M**

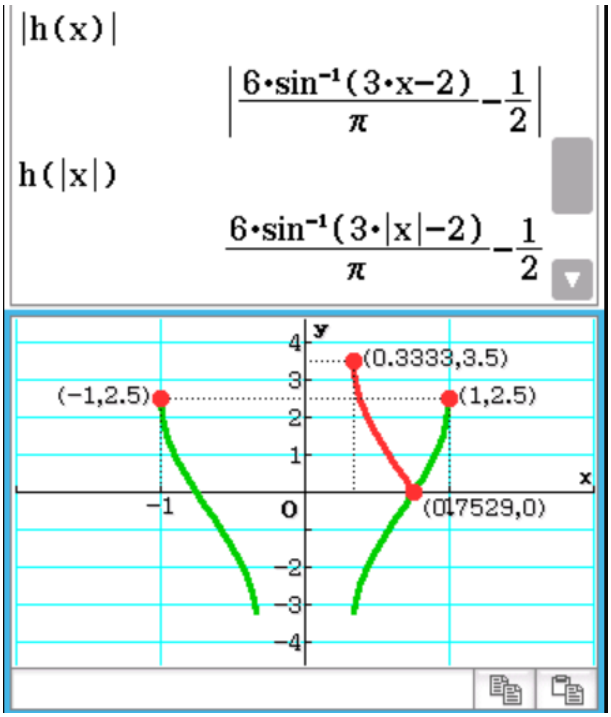
There are no stationary points on the graph of g.



$h(x) = \dfrac{6}{\pi}\arcsin(3x - 2) - \dfrac{1}{2}$ and $y = |h(x)|$ and $y = h(|x|)$

d. Let $|h(x)| = h(|x|)$. Solve graphically. Graphs intersect and overlap for $x \in \left[\dfrac{\sqrt{6} - \sqrt{2} + 8}{12}, 1\right]$

So graphs intersect for infinite values for the interval $x \in \left[\dfrac{\sqrt{6} - \sqrt{2} + 8}{12}, 1\right]$ **1A**



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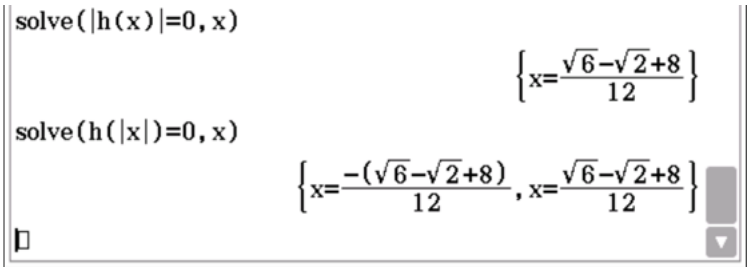
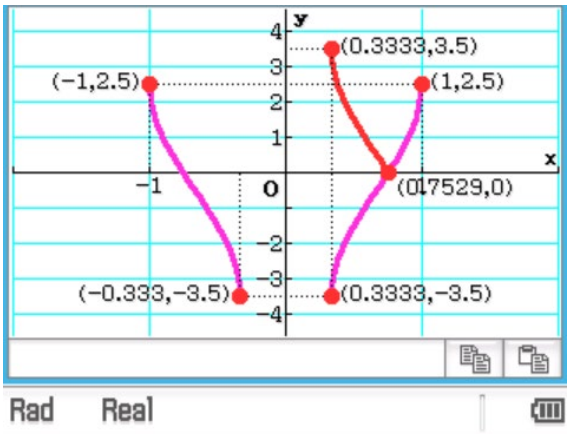
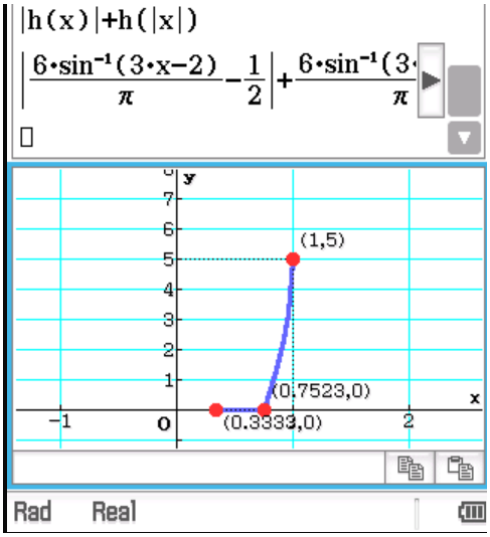


e. $y=|h(x)|+h(|x|)$.

Domain: $\left[\frac{1}{3},1\right]$

Range: $[0,5]$ **1A**





Question 2 (9 marks)

Consider the complex numbers $z_1=\frac{\sqrt{6}-\sqrt{2}}{4}+\frac{\sqrt{6}+\sqrt{2}}{4}i$ and $z_2=\frac{\sqrt{6}+\sqrt{2}}{4}+\frac{\sqrt{6}-\sqrt{2}}{4}i$ where $z\in C$

a. i.

$$|z_1|=\sqrt{\left(\frac{\sqrt{6}-\sqrt{2}}{4}\right)^2+\left(\frac{\sqrt{6}+\sqrt{2}}{4}\right)^2}=1$$

$$|z_2|=\sqrt{\left(\frac{\sqrt{6}+\sqrt{2}}{4}\right)^2+\left(\frac{\sqrt{6}-\sqrt{2}}{4}\right)^2}=1$$

$$z_1=\text{cis}\frac{5\pi}{12}$$

$$z_2=\text{cis}\frac{\pi}{12}$$

1A

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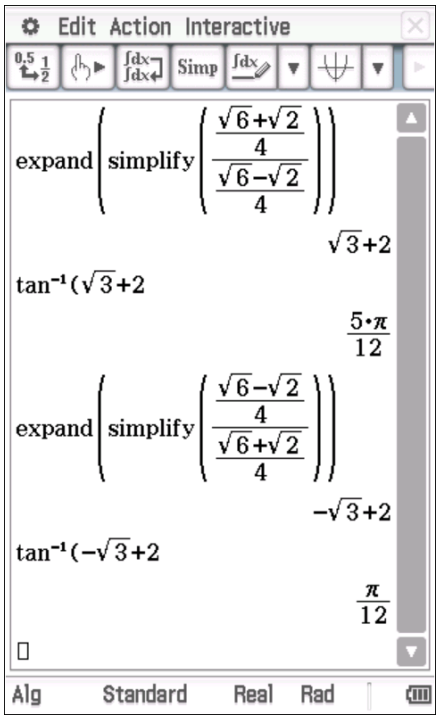

a. ii

$z_1 z_2 = 1^2 \operatorname{cis}\left(\frac{5\pi}{12} + \frac{\pi}{12}\right) = \operatorname{cis}\frac{\pi}{2}$ **1A**

$\frac{z_1}{z_2} = \frac{1}{1}\operatorname{cis}\left(\frac{5\pi}{12} - \frac{\pi}{12}\right) = \operatorname{cis}\frac{\pi}{3}$ **1A**

Given $u = 2\operatorname{cis}\left(\frac{\pi}{2}\right)$ and $v = 2\operatorname{cis}\left(-\frac{2\pi}{3}\right)$.

b. $\{z : z - u = k(v - u), z \in C \text{ and } k \in R\}$ in the form $y = ax + b$

$u = 2i$

$v = 2\operatorname{cis}\left(-\frac{2\pi}{3}\right) = 2\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) = -1 - \sqrt{3}i$ **1M**

$z - u = k(v - u)$

$x + iy - 2i = k\left(-1 - \sqrt{3}i - 2i\right)$

$x = -k$ where x is a variable and k a real constant

$y - 2 = -k\left(\sqrt{3} + 2\right)$

$y - 2 = x\left(\sqrt{3} + 2\right)$

$y = \left(\sqrt{3} + 2\right)x + 2$ **1A**

c. $y = \left(\sqrt{3} + 2\right)x + 2$ with gradient $= \sqrt{3} + 2$ and y-intercept $(0, 2)$

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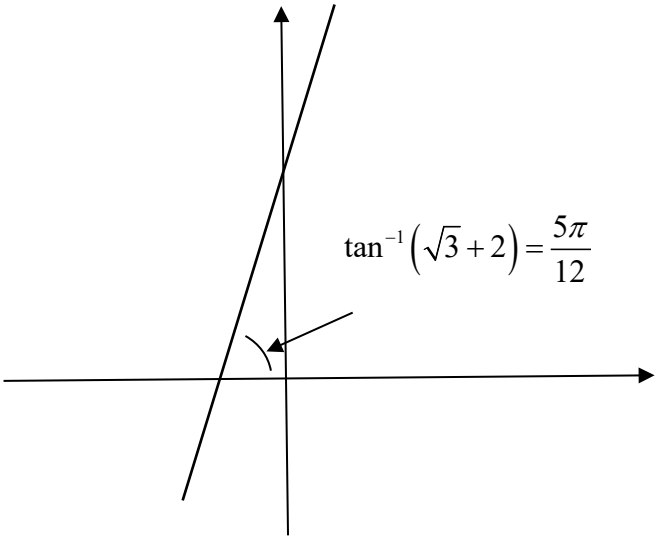
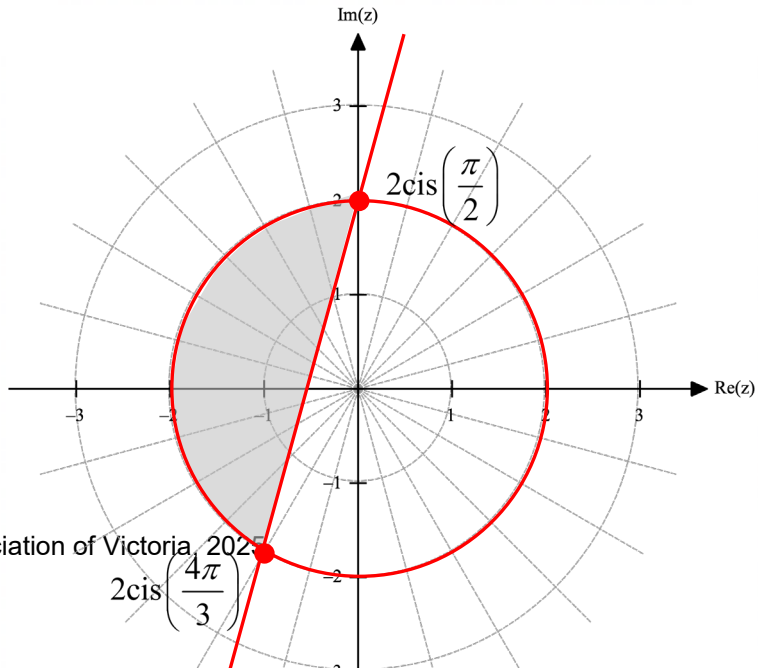


Angle $=\frac{5\pi}{12}$ **1A**

d. Line forms a segment within the circle.

i. Sketch both line and circle, shading minor segment. **2A**



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ii. Area of minor segment

$$\theta = \frac{\pi}{2} + \frac{\pi}{3} = \frac{5\pi}{6}$$

$$\text{Segment area} = \frac{r^2}{2}\left(\theta - \sin(\theta)\right)$$

$$= \frac{2^2}{2}\left(\frac{5\pi}{6} - \sin\left(\frac{5\pi}{6}\right)\right)$$

$$= \frac{5\pi}{3} - 2 \times \frac{1}{2}$$

$$= \frac{5\pi}{3} - 1$$ **1A**

Question 3 (12 marks)

a.

$$\frac{dT}{dt} = kT\left(1 - \frac{T}{40}\right)$$

$$\frac{dT}{dt} = kT\left(\frac{40 - T}{40}\right)$$

$$\frac{dt}{dT} = \frac{1}{kT}\left(\frac{40}{40 - T}\right)$$

$$\therefore t = \frac{1}{k}\int \frac{1}{T}\left(\frac{40}{40 - T}\right)dT$$ **1M**

Partial Fractions

$$\frac{1}{T}\left(\frac{40}{40 - T}\right) = \frac{A}{T} + \frac{B}{40 - T}$$

$$\therefore 40 = A(40 - T) + BT$$

Let $T = 0, 40 = 40A \therefore A = 1$

Let $T = 40, 40 = 40B \therefore B = 1$

Giving $t = \frac{1}{k}\int\left(\frac{1}{T} + \frac{1}{40 - T}\right)dT$ **1M**

$$t = \frac{1}{k}\left(\log_e|T| - \log_e|40 - T|\right) + c$$

$$k(t - c) = \log_e\left|\frac{T}{40 - T}\right|$$

$$e^{k(t-c)} = \frac{T}{40 - T} \therefore e^{k(t-c)}(40 - T) = T \text{ for } 25 \le T < 40$$

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$40e^{k(t-c)} - Te^{k(t-c)} = T$

$T\left(e^{k(t-c)} + 1\right) = 40e^{k(t-c)}$

Giving $T = \dfrac{40e^{k(t-c)}}{e^{k(t-c)} + 1}$

Let $e^{k(t-c)} = Ae^{kt}$ where $A, k \in R$

$T = \dfrac{40Ae^{kt}}{1 + Ae^{kt}}$ as required **1A**

b. $\lim\limits_{t \to \infty} T = \dfrac{40Ae^{kt}}{1 + Ae^{kt}} = 40$ **1A**

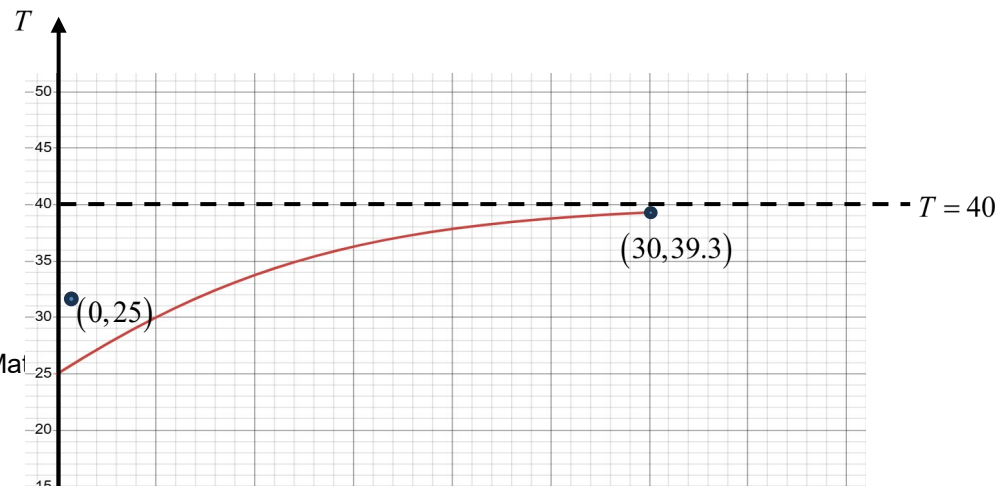
c. Maximum possible temperature approaches 40^0 .
OR horizontal asymptote at $T = 40$ **1A**

d. $T = \dfrac{40Ae^{kt}}{1 + Ae^{kt}}$, using: $t = 0, T = 25$ and $t = 5, T = 30$.

$25 = \dfrac{40Ae^0}{1 + Ae^0} \therefore A = \dfrac{5}{3}$ **1A**

$30 = \dfrac{40\frac{5}{3}e^{5k}}{1 + \frac{5}{3}e^{5k}} \therefore k = \dfrac{1}{5}\log_e\left(\dfrac{9}{5}\right)$ **1A**

e. Sketch the graph of $y = T(t)$ for $t \in [0, 30]$. **2A**



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f.

$$T=\frac{40\left(\frac{5}{3}\right)e^{\frac{1}{5}\log_e\left(\frac{9}{5}\right)t}}{1+\left(\frac{5}{3}\right)e^{\frac{1}{5}\log_e\left(\frac{9}{5}\right)t}}$$

$$T=\frac{40\times\frac{5}{3}\left(\frac{9}{5}\right)^{\frac{t}{5}}}{1+\frac{5}{3}\left(\frac{9}{5}\right)^{\frac{t}{5}}}$$

$$T=\frac{200\left(\frac{9}{5}\right)^{\frac{t}{5}}}{3+5\left(\frac{9}{5}\right)^{\frac{t}{5}}}$$

At $T=36,\quad 36=\dfrac{200\left(\frac{9}{5}\right)^{\frac{t}{5}}}{3+5\left(\frac{9}{5}\right)^{\frac{t}{5}}}\Rightarrow t=14.35$ mins

$\dfrac{dT}{dt}\Big|_{t=14.35}=0.42321$ **1M**

At $T=36,\quad 36=25+n^{\frac{3}{7}}\Rightarrow n=269.1$ persons

$\dfrac{dT}{dn}\Big|_{n=269.1}=0.01752$ **1M**

$$\frac{dT}{dt}=\frac{dT}{dn}\times\frac{dn}{dt}$$

$$\frac{dn}{dt}=\frac{dT}{dt}\div\frac{dT}{dn}=\frac{0.42321}{0.01752}=24.16\approx 24 \text{ persons/min}$$

1A

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```
define f(x)=200×(9/5)^(x/5) / (3+5×(9/5)^(x/5))
                                                    done
define d(x)=d/dx(f(x))
                                                    done
define g(n)=25+n^(3/7)
                                                    done
define h(n)=d/dn(g(n))
                                                    done
solve(g(n)=40,n)
                                        {n=554.8977167}
solve(f(x)=36,x)
                                        {x=14.34533185}
d(x)|x=14.34533185
                                          0.4232063989
solve(g(n)=36,n)
                                         {n=269.101591}
h(n)|n=269.101591
                                          0.0175186096
0.4232063989/0.0175186096
                                           24.15753354
```

Question 4 (10 marks)

a. $\underset{\sim}{r}(s,t)=\underset{\sim}{i}+3\underset{\sim}{k}+s\left(-2\underset{\sim}{i}+3\underset{\sim}{j}+\underset{\sim}{k}\right)+t\left(\underset{\sim}{i}-2\underset{\sim}{j}-\underset{\sim}{k}\right)$

Normal to the plane, $\underset{\sim}{n}=\left(-2\underset{\sim}{i}+3\underset{\sim}{j}+\underset{\sim}{k}\right)\times\left(\underset{\sim}{i}-2\underset{\sim}{j}-\underset{\sim}{k}\right)=-\underset{\sim}{i}-\underset{\sim}{j}+\underset{\sim}{k}$ **1M**

Point on the plane $(1,0,3)$

$d=\underset{\sim}{n}.(1,0,3)=2$

Plane equation, $-x-y+z=2$

$\Rightarrow x+y-z=-2$ **1A**

```
crossP([-2 3 1],[1 -2 -1])
                                        [-1 -1 1]
dotP([-1 -1 1],[1 0 3])
                                                2
```

b. Let, the equation of parallel plane Π_2 is $x+y-z=d_2$

Point $(2,3,1)$ is on plane Π_2. So, $2+3-1=d_2 \Rightarrow d_2=4$

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Equation of plane Π_2 is $x+y-z=4$ **1A**

c. Point on plane Π_1 $A(1,0,3)$

Point on plane Π_2 $B(2,3,1)$

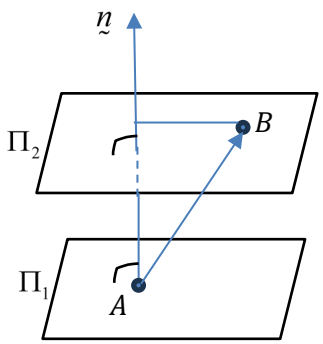
$\vec{AB}=(1,3,-2)$ **1M**

Normal to both planes, $\underset{\sim}{n}=-\underset{\sim}{i}-\underset{\sim}{j}+\underset{\sim}{k}$

Shortest distance between planes Π_1 and Π_2

$=$ scalar projection of $\vec{AB}$ towards $\underset{\sim}{n}$

$=\left|\vec{AB}.\hat{\underset{\sim}{n}}\right|=2\sqrt{3}$ **1A**




```
dotP([1 3 -2],unitV([-1 -1 1]))
                                  -2·√3
```

d. Line equation, $\underset{\sim}{l}(u)=(2+3u)\underset{\sim}{i}+(1-2u)\underset{\sim}{j}+(-1-u)\underset{\sim}{k}$

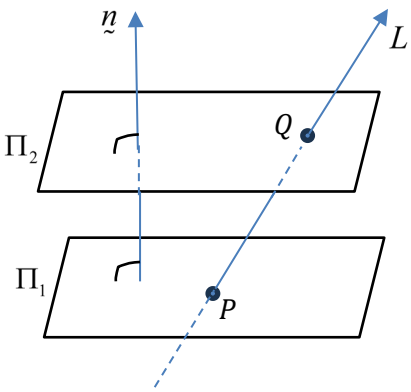
Plane Π_1 equation, $x+y-z=-2$

$\Rightarrow(2+3u)+(1-2u)-(-1-u)=-2$

$\Rightarrow u=-3$

$\underset{\sim}{l}(-3)=-7\underset{\sim}{i}+7\underset{\sim}{j}+2\underset{\sim}{k}$

$\therefore P(-7,7,2)$ **1A**



Plane Π_2 equation, $x+y-z=4$

$\Rightarrow(2+3u)+(1-2u)-(-1-u)=4$

$\Rightarrow u=0$

$\underset{\sim}{l}(0)=2\underset{\sim}{i}+\underset{\sim}{j}-\underset{\sim}{k}$

$\therefore Q(2,1,-1)$ **1A**

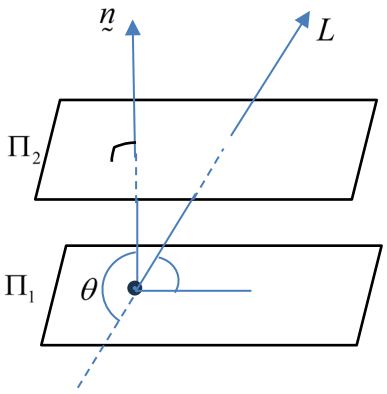
e. Direction vector of line L, $\underset{\sim}{d}=3\underset{\sim}{i}-2\underset{\sim}{j}-\underset{\sim}{k}$

Normal to plane Π_1, $\underset{\sim}{n}=-\underset{\sim}{i}-\underset{\sim}{j}+\underset{\sim}{k}$

Angle between normal and direction vector,

$\cos(\theta)=\dfrac{\underset{\sim}{n}.\underset{\sim}{d}}{|\underset{\sim}{n}||\underset{\sim}{d}|}\Rightarrow\theta=108^o$

Acute angle between the line and plane



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$= 90^o - \left(180^o - 108^o\right) = 18^o$ **1A**

```
angle([3 -2 -1],[-1 -1 1])
                          107.9752838
```

f. Line equation, $\underset{\sim}{l}(u) = 2\underset{\sim}{i} + \underset{\sim}{j} - \underset{\sim}{k} + u\left(3\underset{\sim}{i} - 2\underset{\sim}{j} - \underset{\sim}{k}\right)$

Direction vector of line L, $\underset{\sim}{d} = 3\underset{\sim}{i} - 2\underset{\sim}{j} - \underset{\sim}{k}$

Point on line L, $M(2,1,-1)$

Point on plane Π_3, $N(-5,3,4)$

$\vec{MN} = (-7,2,5)$

Normal vector on plane Π_3, $\underset{\sim}{n} = \vec{MN} \times \underset{\sim}{d} = 8\underset{\sim}{i} + 8\underset{\sim}{j} + 8\underset{\sim}{k} \equiv \underset{\sim}{i} + \underset{\sim}{j} + \underset{\sim}{k}$ **1M**

$\underset{\sim}{n}.(-5,3,4) = 2$

Equation of plane Π_3, $x + y + z = 2$ **1A**

```
crossP([-7 2 5],[3 -2 -1])
                          [8 8 8]
dotP([1 1 1],[-5 3 4])
                                2
```

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Question 5 (9 marks)

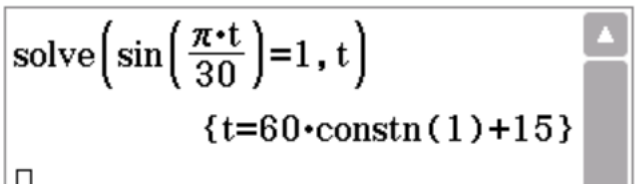
a. $z = 4\sin\left(\frac{\pi t}{30}\right) + 6$

$z_{\max} = 4 \times 1 + 6 = 10\text{m}$ **1A**

b. At $z_{\max}$, $\sin\left(\frac{\pi t}{30}\right) = 1$

$t = 60n + 15, n \in Z^{+} \cup \{0\}$

$\therefore a = 60, b = 15$ **1A**

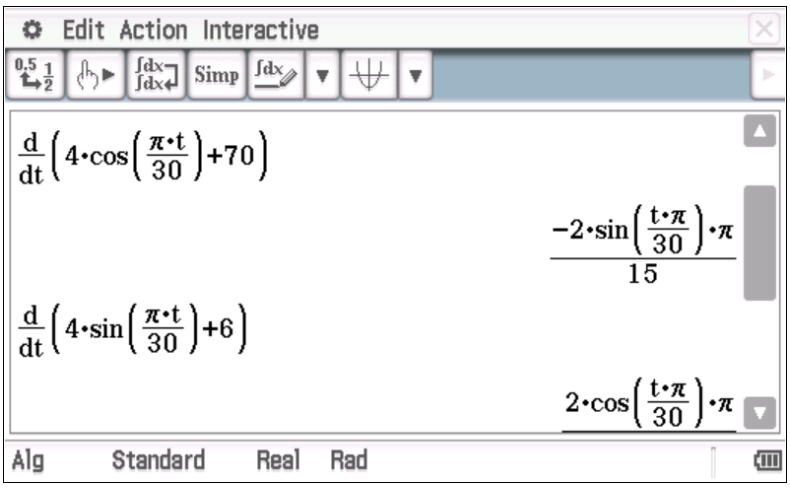


c. $\dot{\underset{\sim}{r}}(t) = \frac{d}{dt}\left(\underset{\sim}{r}(t)\right) = -\frac{2\pi}{15}\sin\left(\frac{\pi t}{30}\right)\underset{\sim}{i} + \frac{2\pi}{15}\cos\left(\frac{\pi t}{30}\right)\underset{\sim}{k}$

$\left|\dot{\underset{\sim}{r}}(t)\right| = \sqrt{\left[-\frac{2\pi}{15}\sin\left(\frac{\pi t}{30}\right)\right]^2 + \left[\frac{2\pi}{15}\cos\left(\frac{\pi t}{30}\right)\right]^2}$ **1M**

$= \frac{2\pi}{15}\sqrt{\left[\sin\left(\frac{\pi t}{30}\right)\right]^2 + \left[\cos\left(\frac{\pi t}{30}\right)\right]^2}$ **1M**

$= \frac{2\pi}{15}$ m/s



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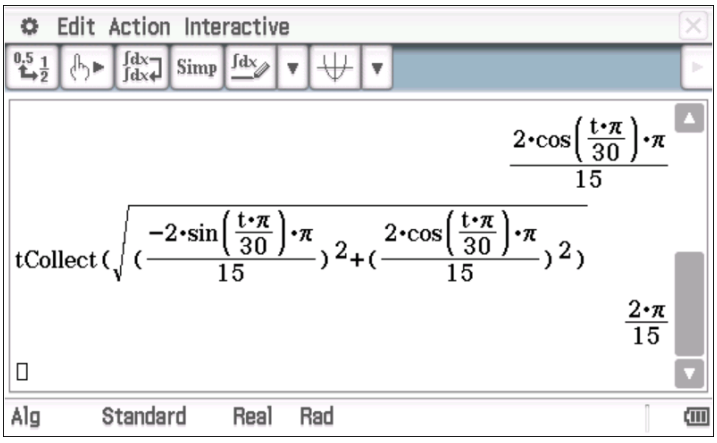
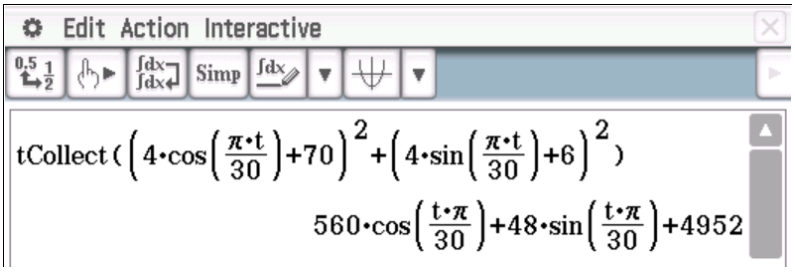


d. $f(t)=\left(4\cos\left(\frac{\pi t}{30}\right)+70\right)^2+\left(4\sin\left(\frac{\pi t}{30}\right)+6\right)^2$

$=560\cos\left(\frac{\pi t}{30}\right)+48\sin\left(\frac{\pi t}{30}\right)+4952$ **1M**

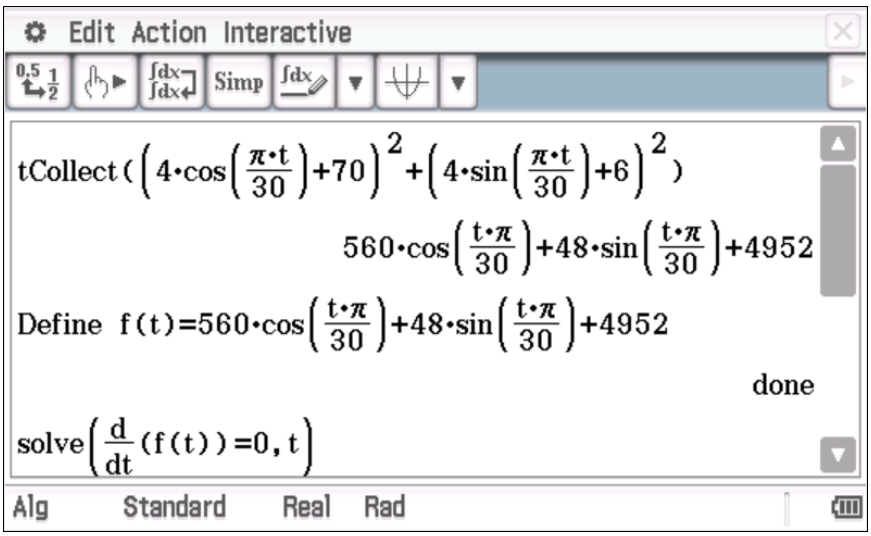


Solve $f'(t)=0$

$t=\frac{30}{\pi}\tan^{-1}\left(\frac{3}{35}\right)+30n$ **1M**

$t=\frac{30}{\pi}\tan^{-1}\left(\frac{3}{30+5}\right)+30n$

$\therefore c=30, d=3$ **1A**



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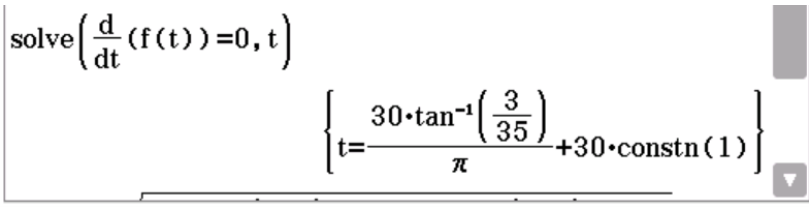


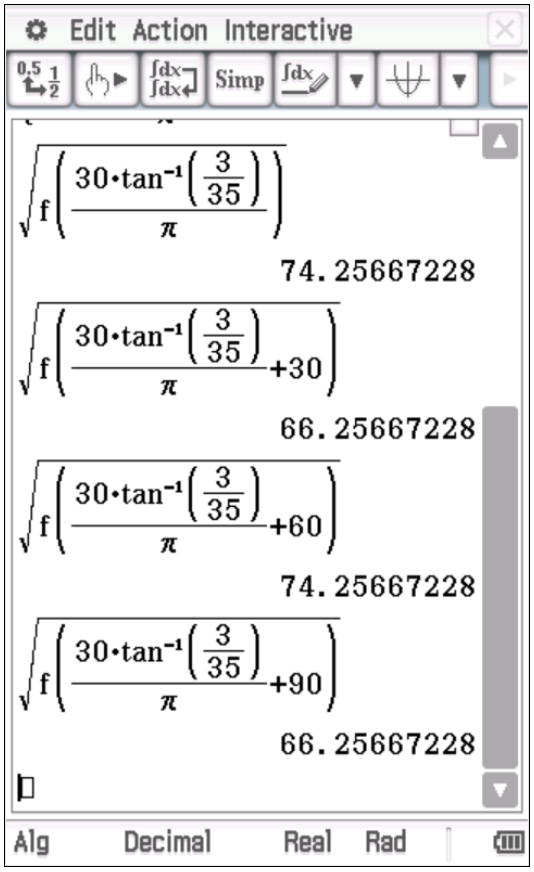
e. For $n=0$, $t=\frac{30}{\pi}\tan^{-1}\left(\frac{3}{35}\right)$

$$\sqrt{f\left(\frac{30}{\pi}\tan^{-1}\left(\frac{3}{35}\right)\right)}=74.3\,\text{m} \qquad \textbf{1A}$$

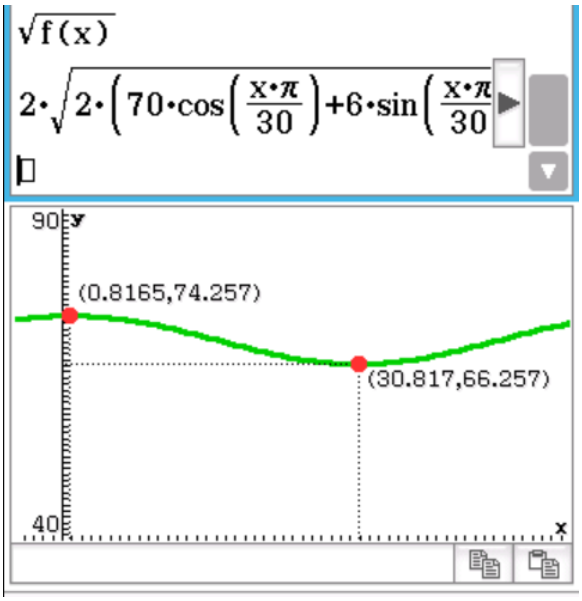
For $n=1$, $t=\frac{30}{\pi}\tan^{-1}\left(\frac{3}{35}\right)+30$

$$\sqrt{f\left(\frac{30}{\pi}\tan^{-1}\left(\frac{3}{30+5}\right)\right)}=66.3\,\text{m} \qquad \textbf{1A}$$

Minimum distance = 66.3 m and maximum distance = 74.3 m



and on graph



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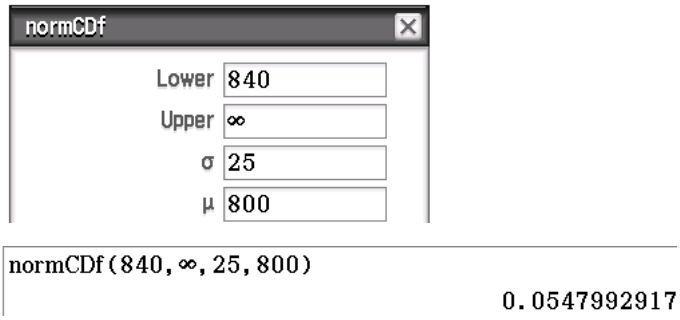
Question 6 (11 marks)

a. $\mathrm{E}(B) = 12 \times 60 + 80 = 800g$

$\mathrm{sd}(B) = \sqrt{12 \times 6^2 + (13.9)^2} = 25g$ **1A**

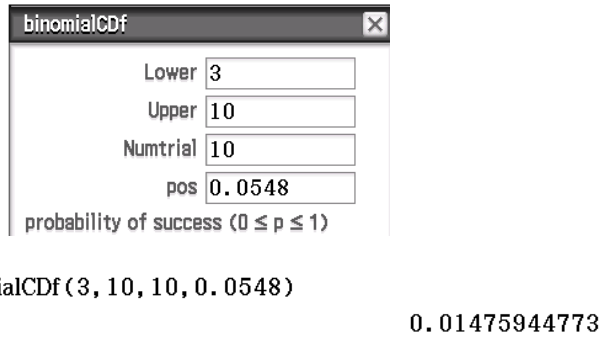
$B \sim \mathrm{N}(800, 25^2)$

$\Pr(B > 840) = 0.0548$ **1A**



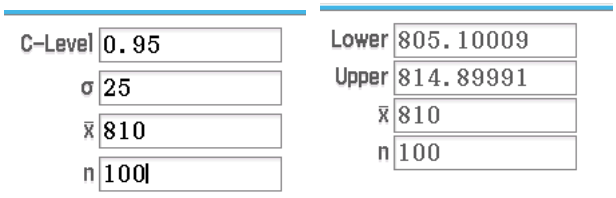
b. Let, $W \sim \mathrm{Bi}(n, p) \sim \mathrm{Bi}(10, 0.0548)$

$\Pr(W \geq 3) = 0.015$ **1A**



c. 95% confidence interval $= (805.1, 814.9)$

1A

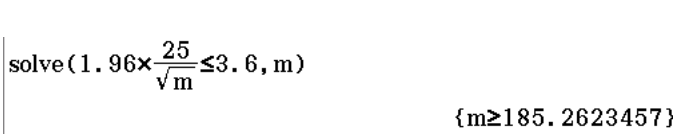


d. Error margin $= 1.96 \times \dfrac{\sigma}{\sqrt{m}} \leq 3.6$

$1.96 \times \dfrac{25}{\sqrt{m}} \leq 3.6$

1M

$m \geq 185.26$



$\therefore m = 186$ **1A**

e. $H_0 : \mu = 60$

$H_1 : \mu > 60$ **1A**

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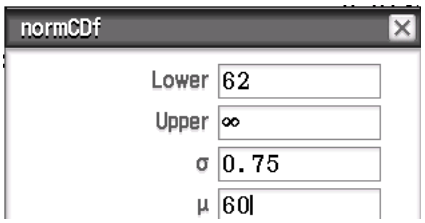
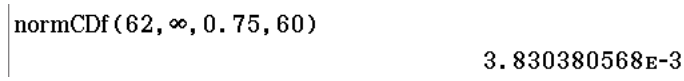
f. Let, X = weight of an egg

$$X \sim \mathrm{N}\left(60, 6^2\right)$$

$$\bar{X} \sim \mathrm{N}\left(60, \left(\frac{6}{\sqrt{64}}\right)^2\right) \sim \mathrm{N}\left(60, (0.75)^2\right)$$



p-value $= \Pr\left(\bar{X} \geq 62 \middle| \mu = 60\right) = 0.0038$ **1A**

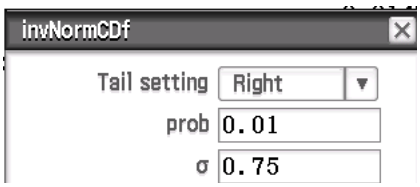


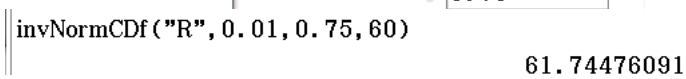
As p-value < 0.01 (significance level), the null hypothesis is rejected and hence test concludes that the new meal increases egg weight. **1A**

g. $\bar{X} \sim \mathrm{N}\left(60, (0.75)^2\right)$

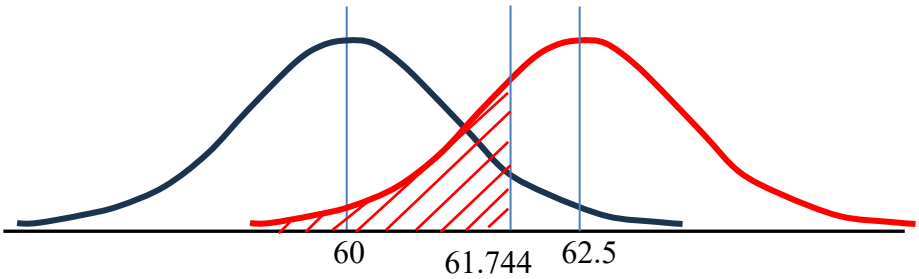
Critical egg weight, $\Pr\left(\bar{X} > c\right) = 0.01$

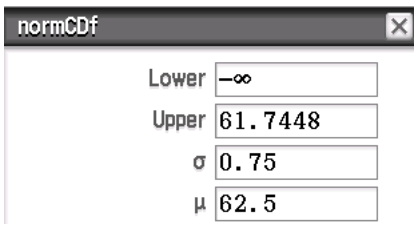
$\Rightarrow c = 61.7448$ **1M**

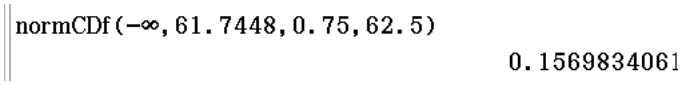




Type II error probability: $\Pr\left(\bar{X} < 61.7448 \middle| \mu = 62.5\right) = 0.157$ **1A**







END OF SOLUTIONS

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